Note Title 2013/05/12

Theme 5. Integration

Unit 5.1 Antiderivatives

Stewart : Section 4.9 pg 344 - 350

Definition A function F is called an antiderivative of f on an interval I if

$$F'(x) = f(x) \text{ for all } x \text{ in } I.$$

Example ① $f(x) = x^3$ Find the antiderivative.

$$F(x) = \frac{1}{4}x^4 \qquad F'(x) = \frac{4}{4}x^3$$

$$F(x) = \frac{1}{4}x^4 + 100$$

Theorem If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x) + C,$$

where C is an arbitrary constant.

Example

① Find a antiderivative of $f(x) = \sin x$

$F(x) = -\cos x$

② Find all antiderivatives of $f(x) = \frac{1}{x}$

$F(x) = \ln|x|$

Since:

$\frac{d}{dx}(\ln x) = \frac{1}{x}$ and $\frac{d}{dx}\left(\ln(-x)\right) = \frac{1}{-x} \cdot (-1) = \frac{1}{x}$

<u>Example</u> Find all antiderivatives of

① $f(x) = x^2 \sin(x^3)$

$$\frac{d}{dx}\left[-\cos(x^3)\right] = \sin(x^3) \cdot 3x^2$$

$$F(x) = -\frac{1}{3}\cos(x^3) + C$$

① $f(x) = e^{4x} \cos e^{4x}$

$$F(x) = \frac{1}{4}\sin(e^{4x}) + C$$

$$\frac{d}{dx}\left[\frac{1}{4}\sin(e^{4x}) + C\right]$$

$$= \frac{1}{4}\cos(e^{4x})\, e^{4x} \cdot 4$$

② $f(x) = \dfrac{\sec^2\sqrt{1-x}}{\sqrt{1-x}}$

$$\frac{d}{dx}\left[\tan\sqrt{1-x}\right] = \sec^2\left(\sqrt{1-x}\right) \cdot \frac{1}{2}(1-x)^{-\frac{1}{2}} \cdot -1$$

$$F(x) = -2\tan\sqrt{1-x} + C$$

Example Find the function f with

$$f'(x) = \frac{1}{3x+1} \quad \text{and} \quad f(3) = 0$$

$$\frac{d}{dx}\left(\ln|3x+1|\right) = \frac{1}{3x+1} \cdot 3$$

$$f(x) = \frac{1}{3}\ln|3x+1| + C$$

$$f(3) = \frac{1}{3}\ln(10) + C = 0 \Rightarrow C = -\frac{1}{3}\ln(10)$$

$$\Rightarrow f(x) = \frac{1}{3} \ln|3x+1| - \frac{1}{3} \ln(10)$$

Example Find the function f with

$$f''(x) = \cos 2x + e^{2x},$$

$$f'(0) = 1 \quad \text{and} \quad f(0) = 2$$

Sol:

$$f'(x) = \frac{1}{2}\sin 2x + \frac{1}{2}e^{2x} + c_1$$

$$f'(0) = \frac{1}{2}\sin 2(0) + \frac{1}{2}e^{0} + c_1 = 1$$

$$\Rightarrow c_1 = 1/2.$$

$$\Rightarrow f'(x) = \frac{1}{2}\sin 2x + \frac{1}{2}e^{2x} + \frac{1}{2}$$

$$f(x) = -\frac{1}{4}\cos 2x + \frac{1}{4}e^{2x} + \frac{1}{2}x + C_2$$

$$f(0) = -\frac{1}{4}\cos 2(0) + \frac{1}{4}e^{2(0)} + \frac{1}{2}(0) + C_2 = 2$$

$$\Rightarrow C_2 = 2$$

$$\Rightarrow f(x) = -\frac{1}{4}\cos 2x + \frac{1}{4}e^{2x} + \frac{1}{2}x + 2$$